UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT
YUNIVESITHI YA FREISTATA

UFS
NATURAL AND AGRICULTURAL SCIENCES
COMPUTER SCIENCE AND INFORMATICS

CSIS3724 - Software Engineering Project

University Smart Parking & Permit Management System (USPMS)

Introduction / General

You have to complete a software engineering project during this course. It is **compulsory to submit** the project. Any work that is submitted late will result in losing a percentage of the total mark for the project. (See the module guide for more details)

You will make use of a team organization similar to the Democratic Team Approach (§ 4.2 in the textbook).

You are allowed to make assumptions for this project as long as the assumptions do not cut down on the minimum requirements as per the project description and submission requirements.

Project Description

The UFS has experienced significant parking challenges over the past few years due to increasing student enrolment, limited parking infrastructure and a partially manual parking administration process.

Current problems include

- Difficulty finding available parking
- Unauthorized parking in restricted zones
- Inefficient permit management
- Congestion during peak academic hours
- Manual parking fine administration
- Lack of real-time parking information

The Free State University has decided to commission a software system to modernize campus parking operations.

Project Overview

The proposed centralized platform should support the following features:

- **Parking permit management**
  The system should support multiple permit types to accommodate different user needs. It enables administrators to review, approve or reject permit applications while maintaining records of permit's validity period, ensuring that only active and authorized permits are in use.
- **Parking zone management**
  The system should enable the effective organization of parking areas by categorizing the zones based on their intended use. Allows administrators to define and manage parking restrictions for each zone, ensuring that parking rules are applied consistently.
- **Visitor parking management**
  To introduce visitor parking management by allowing pre-registration based on visitor type, enabling administrators to search for and reserve parking bays in appropriate zones, generating temporary parking permits and supporting the cancellation of reservations when required.
- **Notifications and alerts**
  Keep users informed by sending updates / notifications for permit expiry, reservation confirmations, etc. helping to improve compliance and ensure users remain aware of important parking related events.

The system will be utilized by students, staff (academic, administrative), campus security and visitors. The application must be easily usable by the various roles/stakeholders. The approach you wish to utilize is completely up to your discretion, but will require a user interface, application and a database. You will design the complete system up until the implementation phase.

For the Software Project Management Plan give special consideration to the requirements of working at a distance. The ability for your team to be able to use a blended approach (i.e. work from home, online and face-to-face) is a requirement. Design a SPMP that specifies exactly how the development team will be supported when programming the application.

Submission requirements

1. Record of hours spent on the project

You should maintain a record of the time (in hours) that you spend on the project. At the end of the project, your information in this regard should be represented as follows:

Requirements	
Specification / Analysis	
Design	

Implementation	
Other	
Total	

Tasks such as documentation, administration, planning and management should also be reflected in one or more of the above categories. You must also indicate what kind of tasks you reflected under the "Other" column.

Chapters 1, 2 and 3 will provide background for this task
This record should be handed in with the final submission

2. Specification document

You have to compile a specification document that could serve a real specification document between you and the client. You should make use of UML (refer to Chapter 13 to see which techniques are relevant). Note that you only have to provide a UML design for the business layer (business objects) of the problem and not for the interface layer or the data access layer. The specification document should be handed in on **the date supplied by the lecturer**. The client (lecturer) will take a look at it and give feedback. The feedback could entail changes to the planned systems (and the document). The feedback must be incorporated into the final version.
The initial specification document and the final version of the specification document must again be handed in with the final system. Add an appendix to the final version that briefly lists the changes that were made to the final version. Please note that the initial version of the document may have a higher mark allocation than the final version.

Chapters 1, 2, 3, 4, 11 and 13 will provide background for this task.

3. Software project management plan (SPMP)

Once the specifications have been completed, you will have to draw up a software project management plan. Please refer to Sections 9.4, 9.5 and Appendix F for examples. Although these will give you an idea of the layout of the SPMP, it is important that your SPMP reflects your own circumstances (that is, this project's detail, your own resources, planning of the schedule according to your real-life situation, etc.)

You have to add all the sections in the example SPMP to your own SPMP and add relevant information to them as far as possible. If there is a section that you feel is totally inappropriate to your SPMP, you could still include the corresponding heading, but indicate that it is "Not applicable"

You have to employ an Iterative-and-Incremental Model for the development. Ensure that you apply what you have learned about development operations and incorporate these techniques into your SPMP.

The estimates for the duration of the project should be done in hours and not in

weeks as in Sections 1.1 and 5.2 in the example in Appendix F. You should project the cost of the project as if you had to finish the whole of the project; that is, also include aspects like testing, etc.

Please note that you do not have to draw up (or hand in) a test plan.

4. Quotation for the system

You must also hand in a quotation together with the specification document. Use the following guidelines to determine the quotation amount:
Each team member works at R350 per hour.
Provide for general expenses (water and electricity, petrol, etc.) at 5% of the total amount of the man-hours.
Include VAT (15%)

NB: If the number of estimated hours differs substantially from the hours that actually went into the system, it will not necessarily have a negative effect on your marks as long as you can explain the difference.

Chapter 1, 2, 3 and 4 will provide background for this task

5. Overview of development effort

Together with the final hand-in, you must also hand in a concise description of how you applied the Iterative-and-Incremental Model. Would you do anything differently if you were to develop a similar product?

Chapters 2 and 4 will provide background for this task.

6. Design documentation

You have to provide an object-oriented design for the business layer of the project. Refer to Chapter 14 for more detail. Please note that you should only provide detailed design for complex calculations and algorithms that are required.

Although your design need not reflect the interface layer and the data access layer, you must also hand in a brief description (half a page) of how the business objects “link” with the interface layer and the data access layer.

Your design document should also state which technologies you have decided to use and whether you are developing an online or Windows based system. Motivation should also be provided for your choices.

The design document should be handed in with the final system.

Chapters 1, 2, 3, 4. And 14 will provide background for this task.

7. General information

What you put into this module will reflect what you get out of it. The quality of your project will determine your semester mark to a large extent. Assignments are to be submitted on time, and in the correct format.

Where:
Submissions must be handed in on Blackboard. You will be required to declare that the work submitted is your own.

Format:
All assignments must be typed up, and all diagrams must be completed using MS Visio or Draw.IO in UML. All submissions must adhere to the following

- All submissions must be in Microsoft Word docx, no other formats will be accepted
- Use the font Arial in size 10 as the body text
- Filename should be your Surname(StudentNumber)_GroupNumber, followed by the assignment number. (e.g. Doe(2020123458)_G3_Portfolio). Ensure to list all team members' information on the title page
- Each submission must have a cover page with the team information (i.e. full name and student number of all team members)

Draft submission, feedback and final copy submission process:
The due dates of the drafts and final copies of assignments will be made available on Blackboard.
The process is as follows:

- Your *draft* submission will not be reviewed, and constructive feedback will be provided in class (no marking!)
- You will then have an opportunity to improve your draft submission and submit a *final* copy
    - Marking will be guided by the marking rubric provided
- A final version of all projects must be handed together and form your final portfolio
- **Failure to submit a final portfolio will result in an incomplete for CSIS3724**

The whole project should be attempted in a way that helps to make any further maintenance easy.

All documentation must be handed in as a Microsoft docx document on Blackboard. Each document will have a cover page with full name (including group number), table of contents, student name and numbers, along with page numbers.

Marks will also be awarded for your "professionalism". This includes aspects such as the management of the project and meeting deadlines.